MATH 6241- Stochastic Processes

Reviews probability theory; conditional probability and expectations; discrete-time and continuous time Markov chains; Poisson process and exponential distributions; (renewal theory and its applications; queueing theory) Brownian motion; stochastic systems. Applications to Monte Carlo simulation and various real-world scenarios. Involve practical applications through computer simulations by a language programming or/and project in real-world problems, such as finance, biostatistics, machine learning, etc.

Instructor: He Wang

Email: **he.wang@northeastern.edu** (mailto:he.wang@northeastern.edu)

Website: **https://hewang.sites.northeastern.edu/** (https://hewang.sites.northeastern.edu/)

Github: **https://github.com/wanghemath** (https://github.com/wanghemath)

I am organizing the **Northeastern Applied and Interdisciplinary Mathematics (AIM) Seminar.** (https://aimath.sites.northeastern.edu/)

I also organize the **Data Club** (https://aimc.sites.northeastern.edu/) for MS students in math department.(**information** (https://northeastern.instructure.com/courses/245365/pages/data-club-information))

Class time and room:

Class 3:25 pm - 5:05 pm Tuesday Richards Hall 140

The class is in-person and on-ground.

Student Feedback:

- Feel free to **email** me anytime in the semester if you would like to give feedback to me for course improvement. I appreciate your feedback.
- In the end of the semester, you also need to complete the **TRACE** survey for the course for university administration purposes. **https://northeastern.bluera.com/** (https://northeastern.bluera.com/)

Lecture Notes and supporting materials are under Modules (https://northeastern.instructure.com/courses/245365/modules)

Class notes, Solutions to Homework, labs, and practices are:

Class notes (https://northeastern.instructure.com/courses/245365/pages/class-notes)

Assignment Solutions (https://northeastern.instructure.com/courses/245365/pages/assignment-solutions)

Practice Questions (https://northeastern.instructure.com/courses/245365/pages/practice-questions)